



A Design for a Secure Network of Networks Using a Hardware and Software Co-Engineering Architecture

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ABSTRACT

This work concerns the demonstration of a security solution for a network of networks, which comprises heterogeneous devices and utilises diverse communication protocols. The security solution used in this work employs an architecture presented in a previous work, which is based upon the concept of hardware and software security co-engineering.

1 INTRODUCTION

As the Internet of Things (IoT) continues to expand, individual systems of (sub)systems and networks of (sub)networks, such as smart homes, smart vehicles, and smart grids, are more and more often integrated into broader systems and networks, e.g., in the context of a smart city. This provides attackers with more opportunities, as, in this case, malware can spread much more easily from one network to the other, potentially leading to significant losses and damages [2].

Establishing security in this case is not trivial, due to the highly diverse nature of the systems and networks being used, which may be significantly limited in terms of resources and bandwidth. Therefore, a lightweight and cost-efficient solution is required, which also has to be highly scalable and flexible, as networks can be composed of just a few devices or of several hundreds. This work¹ concerns the application

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to a network of networks of such a security solution based upon a hardware and software co-engineering architecture that has been discussed in a previous work [3].

2 THE DEMONSTRATOR

Our demonstrator is based upon an extended version of the system of systems presented in [3], which consists of heterogeneous devices connected using a variety of communication protocols. A proof-of-concept security solution for the resulting network of diverse networks is briefly described and applied to this network of networks.

2.1 The Demonstration Setup

The demonstrator consists of various evaluation boards that are often employed in the context of the IoT, connected in a network of networks utilising a number of different communication protocols, as shown in Figure 1. These devices are also connected to different sensors and actuators, and the overall network employs both wired communication, such as Ethernet, Serial, GPIO, and CAN connections, as well as wireless communication, based on the Wi-Fi, Bluetooth, and LoRaWAN protocols. In this way, different types of systems of systems and networks of networks can be emulated, such as autonomous cars, smart homes, smart grids, or even smart cities consolidating instances of different smart systems and their infrastructure. The network setup is currently based on the master-slave principle, however, other network topologies and layouts can be easily adopted.

2.2 Securing the Network of Networks

The security solution utilises the security architecture described in [3], which is based on hardware and software co-engineering. In particular, a hardware security primitive (a DRAM-based physical unclonable function) serves as the

Jan Philipp Dahms, Mark Rothermel, and Tobias Stensbeck, for their work relevant to this demonstrator, in their role as students of TU Darmstadt.

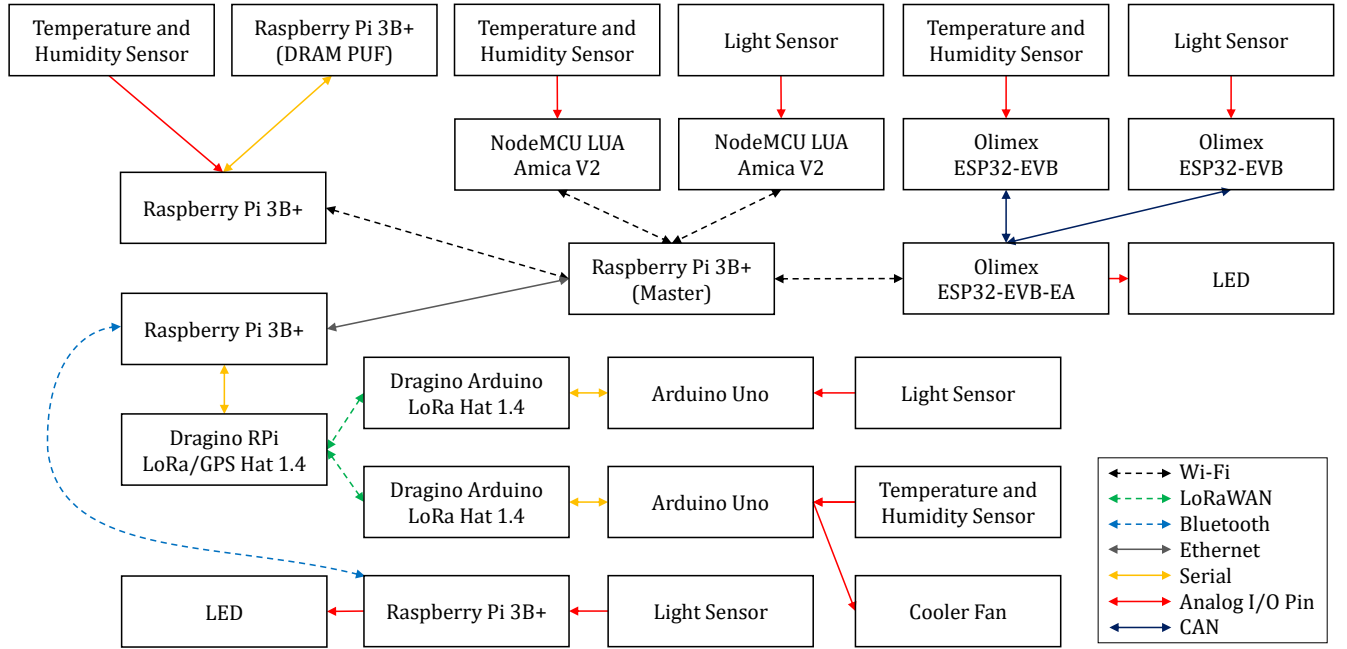


Figure 1: Overview of the demonstrator, which consists of diverse devices and network protocols.

security anchor, providing secure keys for the software algorithms employed for cryptography, namely HMAC-SHA-256 for message authentication, and AES-128-CBC for message encryption. The Message Queuing Telemetry Transport (MQTT) protocol is employed for communication, in order to allow messages to only reach devices that are subscribed to particular “topic”-based channels. Gateway devices are used to allow the MQTT protocol to be ported to network protocols for which it is not natively supported.

Authenticated time messages are exchanged between the master device (time broker) and the slave (client) devices to synchronise the whole network of networks. Nonces are used in order to prevent replay attacks. Synchronised devices can then exchange encrypted data messages, which are also authenticated and contain valid timestamps. The relevant

protocol is shown in Figure 2, while the overall security solution is examined in detail in [3], where it is proven to be scalable, as well as effective against a Dolev-Yao attacker model [1] as it can provide an acceptable level of protection against attacks that target the confidentiality, the integrity, or the availability of the overall network or its subnetworks. An external time server can also act as the time broker, and, apart from client-to-broker data communication, client-to-client and broker-to-client data messages can also be exchanged.

3 DEMO PERFORMANCE EVALUATION

In order to examine the overhead of the security solution, we have measured the overall time T_{msg} required for a message to be encoded, sent, and then be processed by its receiver, as well as for a response to be transmitted, i.e. the sum of the Round Trip Time (RTT) and the Packet Processing Time (PPT). Ethernet has an average T_{msg} of 11.0423 ms, Wi-Fi of 15.4035 ms, LoRaWAN of 60.3316 ms, and Bluetooth of 100.5771 ms. However, Ethernet has a time variance of 20.7333 ms, Wi-Fi of 83.5339 ms, LoRaWAN of 1.1603 ms, and Bluetooth of 58.3505 ms. Thus, Wi-Fi and Ethernet have a T_{msg} suitable for real-time communication, but LoRaWAN can provide the most stable transmission. No up-to-date version is available for the libraries used for the Bluetooth and LoRaWAN connections, which can certainly be optimized, e.g., by supporting the BlueZ 5 driver. Also, a study of the power consumption remains as future work. In general, however, our security solution can be considered as lightweight, cost-efficient, scalable, and flexible, and therefore as a practical solution for protecting networks of networks.

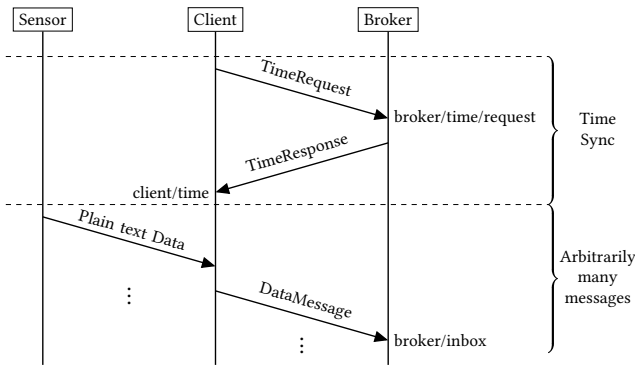


Figure 2: An overview of the proposed security protocol. (Modified from [3].)

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